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RESEARCH LETTERS

Development and Validation of an Integrative Risk Score for Future Risk of Crohn's Disease in Healthy First-Degree Relatives: A Multicenter Prospective Cohort Study

Despite advances in medical therapy, approximately one-half of patients with Crohn's disease (CD) develop complications ultimately requiring surgery. Currently, no treatment offers a chance of cure.[1,2] The increasing incidence and prevalence of CD globally pose a substantial burden at the population and individual levels. Thus, a better understanding of the early triggers of CD is desperately needed to enhance early detection, improve medical therapy, and offer a possibility of interventions that prevent development of CD.

Previous studies have reported candidate predisease biomarkers associated with future development of CD, supporting the concept of a preclinical phase.[3–6] However, these studies lack integration of individual predisease signatures and the multidimensional mechanisms of disease pathogenesis. To date, no risk-stratification model has combined multiple predictive biomarkers to predict future risk of developing CD in healthy at-risk individuals.

In this study, we leveraged a large prospective multicenter cohort of healthy first-degree relatives (FDRs) of people with CD—the Crohn's and Colitis Canada-Genetic, Environmental, Microbial (GEM) study. Using available demographic characteristics, physiologic biomarkers, and fecal microbiome data, we developed an integrative multidimensional risk model—the GEM integrative risk score (IRS)—that predicts an FDR's future risk of developing CD. We subsequently validated the performance of GEM-IRS in 2 testing cohorts.

We included participants with a minimum 3 months of follow-up duration as of February 28, 2021; participants recruited from countries with sufficient median follow-up duration of at least 5 years (ie, Canada, United States, and Israel); and participants with a complete set of blood, urine, and stool samples at recruitment. Fecal microbial profiling was performed by 16S ribosomal RNA sequencing; microbial function was imputed using PICRUSt2.[6,7] Fecal calprotectin (FCP) was measured as a biomarker of gut inflammation and intestinal barrier function was assessed by the ratio of the urinary fractional excretion of lactulose over that of mannitol (LMR).[4]

In total, 3270 healthy FDRs were included: 2713 from North America were randomly split 50:50 into 1356 and 1357 in the North American training and testing cohorts, respectively, and 557 were included from Israel. After quality control of microbiome data and removal of missing data, there were 2619 participants (1170, 1141, and 308, respectively), which we used for subsequent analyses. During a median follow-up duration of 6.8 years (interquartile range 4.2–9.6 years), 61 of 2619 participants (2.3%) developed CD. The baseline demographic characteristics can be found in Supplementary Table 1.

The GEM-IRS, built with random survival forest,[8] resulted in a predictive performance measured by a Harrell's concordance index (Supplementary Material) of 0.951 (95% CI, 0.925–0.976) in the North American training cohort. On validation, the GEM-IRS had a concordance index of 0.789 (95% CI, 0.713–0.865) in the pooled testing cohort. Of note, the performance was higher compared with the other models tested (ie, 0.777 for gradient boosting survival and 0.717 for penalized Cox proportional hazards model).

The cumulative incidence of CD in the 2 testing cohorts was assessed using the quartiles of the GEM-IRS from the training cohort (Figure 1). The fourth quartile compared with the lower 3 quartiles showed a significantly increased risk of CD onset (hazard ratio, 6.42; 95% CI, 3.10–13.30) in the pooled testing cohort.

In the pooled testing cohort, the GEM-IRS had an unadjusted hazard ratio of 2.69 per SD increase (95% CI, 2.06–3.53). After adjusting for each prespecified covariate (ie, CD polygenic risk score, Jewish ethnicity, White race, current smoker, ever smoker, and breastfeeding), the adjusted hazard ratio of the GEM-IRS remained consistent between 2.59 and 2.72 (95% CIs between 1.91 and 3.60) (Supplementary Figure 1).

In predefined subgroups of the pooled testing cohort, the GEM-IRS remained significantly associated with the onset of CD even in FDRs with baseline FCP < 50 μg/g, FCP < 100 μg/g, or LMR ≤ 0.025. In addition, the signal with GEM-IRS remained significant even in participants recruited more than 7 years before diagnosis or last follow-up date (Supplementary Figure 1). Further sensitivity analyses imputing missing values (R package missForest, version 1.4) showed consistent model performance (data not shown).

To gain further insight into the contribution of individual biomarkers, we estimated feature importance; FCP and LMR were the strongest contributors to the model, followed by microbial composition and functional pathways. Network analysis revealed complex interactions between the microbial composition and functions with FCP and LMR. In addition, heatmap and dendrogram analyses identified clusters of microbial composition and functions that were associated with increased or decreased risk of CD (Figure 1 and Supplementary Material).

Abbreviations used in this paper: CD, Crohn's disease; FCP, fecal calprotectin; FDR, first-degree relative; GEM, Genetic; Environmental, Microbial; IRS, integrative risk score; LMR, ratio of the urinary fractional excretion of lactulose over that of mannitol.

Most current article

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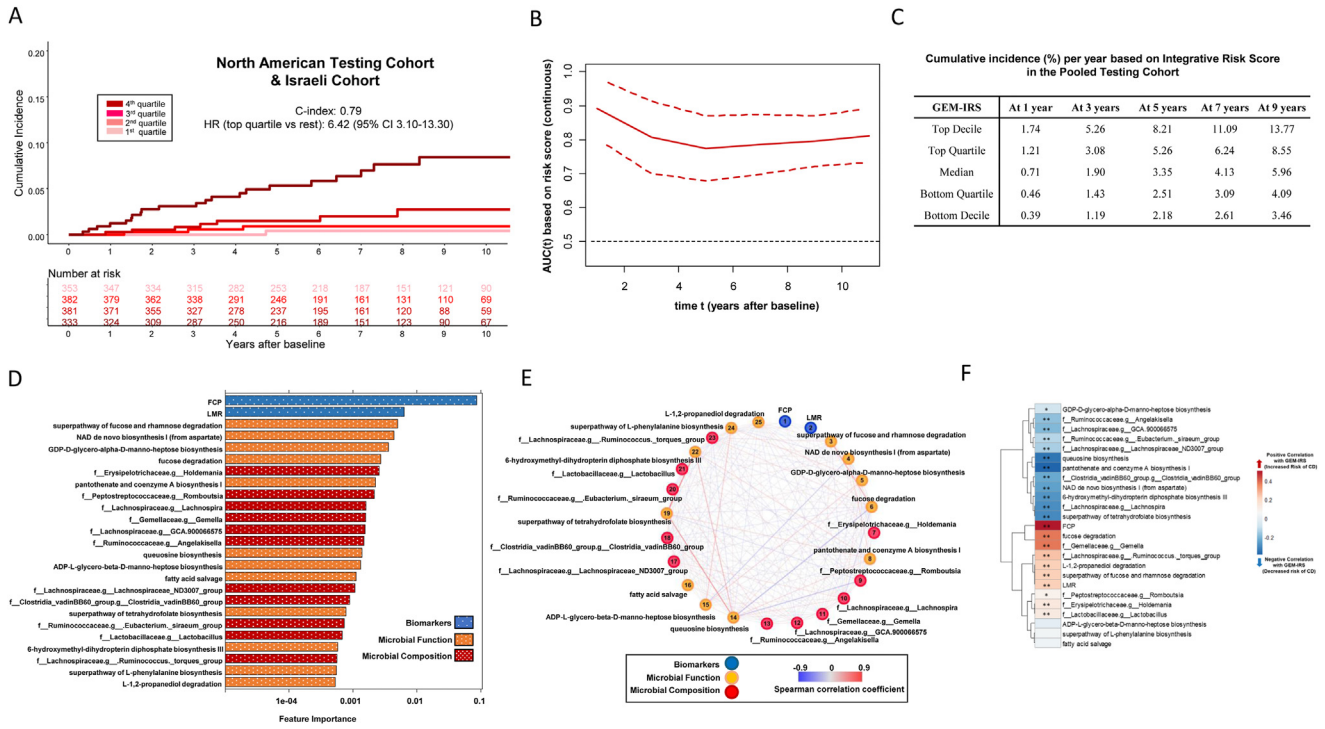


Figure 1. Results of the GEM-IRS. (*A*) Cumulative incidence of CD diagnosis in the pooled testing cohort (North American testing cohort and Israel cohort) using the quartile levels of the GEM-IRS from the training cohort. The fourth quartile (compared with first to third quartiles) had a significantly increased risk of CD onset in pooled testing cohort (hazard ratio [HR], 6.42; 95% CI, 3.10–13.30). (*B*) Time-dependent area under the receiver operating characteristic (AUROC) and 95% CI, based on the GEM-IRS in the pooled testing cohort (R package timeROC). (*C*) Cumulative incidence (%) per year based on the multiple thresholds of the GEM-IRS in the pooled testing cohort. (*D*) Feature importance of the top 25 important biological features from the GEM-IRS trained on the North American training cohort. (*E*) Network analysis based on Spearman correlation among the top 25 important biological features of the GEM-IRS in North American training cohort. The color of the *line* reflects the direction of correlation (*red:* positive correlation; *blue:* negative correlation) and the width of the *line* reflects the strength of correlation. The *blue nodes* represent the biomarkers. *Orange* and *red nodes* represent microbial functions and composition, respectively. (*F*) Heatmap analysis based on Spearman correlation coefficient between the top 25 important biological markers vs GEM-IRS from the training cohort. *Dendrogram* shows clusters generated by the farthest neighbor clustering approach. $^{*}P < .05$; $^{**}P < .005$ for the Spearman correlation analysis. f__, family level; g__, genus level.

To date, this is the first description of an integrative risk model that combines physiological biomarkers of gut inflammation and gut barrier function with fecal microbial composition and functional pathways to assess the future risk of CD among healthy FDRs. The GEM-IRS showed competitive performance in the testing cohorts from 2 different geographic locations (North America and Israel) and provided the relative risk and cumulative incidence based on individual's risk scores.

In the subset of FDRs with baseline FCP < 50 μg/g or LMR < 0.025, using stringent cutoffs,[4,9] we speculated that other factors, for example, gut microbiome composition and functions, could be early events contributing to the risk of developing CD. In our model, the genera *Holdemania*, *Gemella*, and *Ruminococcus torques* group were associated with an increased risk of CD, whereas *Anaerosacchariphilus* species, *Angelakisella*, and *Lachnospira* were associated with a decreased risk. Interestingly, fucose degradation-related microbial pathways were identified as strongly associated with an increased risk of CD. Indeed, previous gene-wide association studies linked an increased risk of CD with *FUT2*-null alleles, supporting the role of fucosylation in protection from CD.[10]

We believe the integrative model showed the potential to risk-stratify high-risk individuals among healthy FDRs, which is the first step toward designing a primary prevention trial in CD. Further research is warranted to develop a model that is clinically applicable; we also need to validate and calibrate the findings in external cohorts (given the possible model instability due to a small number of events) and to determine to what extent asymptomatic persons identified to be at high risk of CD will be willing to undertake interventional therapy. Our results also point toward changes in biomarkers that might direct specific intervention strategies.

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CROHN'S AND COLITIS CANADA-GENETIC, ENVIRONMENTAL, MICROBIAL (CCC-GEM) PROJECT RESEARCH CONSORTIUM

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at https://doi.org/10.1053/j.gastro.2024.08.021.

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Ken Croitoru, MDCM (Conceptualization: Lead; Data curation: Lead; Formal analysis: Equal; Funding acquisition: Lead; Methodology: Lead; Project administration: Lead; Resources: Lead; Supervision: Lead; Validation: Lead; Visualization: Lead; Writing – review & editing: Lead)

Conflicts of interest

The authors disclose no conflicts.

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Data Availability

Requests for raw and analyzed data should follow the instructions provided at http://www.gemproject.ca/data-access/. All submissions will be reviewed by the GEM Project Operating Committee to ensure that the requested samples/data will not interfere in any way with the intended GEM Project analysis of the nested cohort as per the original GEM Project Study Design and is not a duplication of analysis already ongoing. Those proposals meeting this evaluation will be distributed to all members of the GEM Project Steering Committee for review and open discussion. This review will focus on the global scientific merit of the proposal. This review will assess the basic scientific merit and availability of requested samples and data, ensuring there is no compromise of the original intent of the GEM Project. It would be of value to contact a member of the Steering Committee who could help sponsor your application. Those projects achieving a majority vote of approval at the GEM Project Steering Committee will be informed that the GEM Project will provide a letter of support stating that the requested samples or data will be made available to the applicants once the applicant receives funding from a granting agency that applies an independent peer review process to the proposal. The criteria to be used for review of all submissions will include the "scientific relevance" of the proposal and the judged availability of biological material requested. The budget to be requested from a funding agency must allow for any expenses in processing samples or in setting up the appropriate queries of the database. The intent is to allow sufficient time for applicants to consider submission for funding opportunities.

Supplementary Materials and Methods

Study Design

Details of the CCC-GEM study can be found in our earlier article.[6]

Study Outcome and the Prespecified Baseline Variables of Interest

The primary outcome was the time to diagnosis of CD. The time in the study was defined as the number of years from the date of recruitment to the date of CD diagnosis (for FDRs who developed CD) or the date of last follow-up (for FDRs who remained healthy) based on the follow-up data available as of February 28, 2021. Of note, participants who developed ulcerative colitis were removed from the analysis. The prespecified baseline variables included in the subsequent risk-stratification model analysis were as follows: age, sex, relation to proband (sibling or offspring), CD-multiplex family status (having more than 1 FDR with a diagnosis of CD at recruitment), FCP, LMR, Shannon α-diversity index, total number of 16S ribosomal RNA reads, relative abundance of taxa at the genus level of taxonomy, and relative abundance of imputed microbial functional pathways.

Briefly, fecal microbial profiling was performed by sequencing the V4 hypervariable region of bacterial 16S ribosomal RNA gene, as reported previously.[6] We summarized the Amplicon Sequence Variant table into taxonomic ranks using QIIME2. Samples with <10,000 reads were excluded from the analysis. In addition, genera present in <20% of the samples were excluded for subsequent analysis. The taxonomic table was converted into relative abundance values by dividing the read counts of each taxon by the sum of the read counts of a sample at the genus taxonomic level. Microbial function was imputed using the PICRUSt2 platform.[7] We normalized the output abundances of each function by the 16S Amplicon Sequence Variant counts by the picrust2_pipeline.py script. We generated the pathway abundances against the MetaCyc pathway database. Pathways present in <20% of the samples were removed from the pathway table.

Statistical Analysis

To develop and validate a model, the subjects recruited from North America (Canada and United States) were randomly split 50:50 into the North American training and testing cohorts. The GEM-IRS was trained in the North American training cohort and tested in the North American testing cohort and in the Israeli cohort, both pooled and separately. The data from the 2 testing cohorts were never used in the model training procedure and were kept separate to avoid data leakage. Validation in the Israeli cohort was considered to assess the generalizability of the score across geographical regions.

The GEM-IRS was developed using the random survival forest (Scikit-survival 0.16.1 module of the scikit-learn python package [python, version 3.6.1]), a meta-estimator that fits survival trees on subsamples of the dataset and averages the results to control for overfitting and integrates baseline parameters into a model that predicts time to event (further detailed in the following section).[8] The prespecified baseline variables (listed above) were included as predictors in the model. Of note, given the high dimensionality of the microbiome data, we excluded microbial genera and microbial functional pathways with a global test $P > .2$, using Cox proportional hazards regression in the training set. Harrell's concordance index (C-index) was used to assess the model performance, which measures the probability that the predicted event of 2 randomly selected subjects maintaining the same relative order as their true event times.[e1]

Feature importance, a measure of each feature's contribution in the model, was assessed via a permutation-based approach (python package Eli5, version 0.11.0) based on the decrease of C-index after excluding each single feature from the model. A network analysis of the top 25 important biological features based on Spearman correlation was performed (Cytoscape, version 3.8.0) with the intent to identify the complex interactions between the features captured by GEM-IRS. Lastly, a heatmap analysis with cluster dendrogram based on Spearman correlation (R package pheatmap, version 1.0.12) was performed to identify clusters of important features correlating with the GEM-IRS.

The predictive performance of GEM-IRS was validated using the 2 testing cohorts (presented as pooled and as individual cohorts). The cumulative incidence over time ranging up to 10 years according to each quartile of the GEM-IRS in the testing cohorts were presented. To allow comparability across cohorts, the quartile cutoff levels of the GEM-IRS were determined from the training cohort.

A Cox proportional hazards regression was used to estimate the hazard ratio of GEM-IRS on time to CD diagnosis in the testing cohorts. In addition, the hazard ratio was adjusted for predefined risk factors for CD based on previous literature (ie, CD polygenic risk score,[e2] self-declared White race and Jewish ethnicity,[e3,e4] current smoker, ever smoker,[e5] and breastfeeding history[e6,e7]). We further estimated the hazard ratio of the GEM-IRS in prespecified subgroups: participants with no intestinal inflammation (ie, FCP $\leq$ 100 μg/g, FCP $\leq$ 50 μg/g); participants with no abnormal gut barrier function (LMR $\leq$ 0.025)[4,9]; and participants based on follow-up duration (0.5 year and up to 7 years before diagnosis or last follow-up date).

A 2-sided P value $< .05$ was regarded as statistically significant. All analyses were performed using R software, version 3.6.2, unless stated otherwise.

Random Survival Forest

We used random survival forest (scikit-survival, version 0.15.1 module of scikit-learn python package [python, version 3.6.1]; https://scikit-survival.readthedocs.io/en/latest/). The random survival forest consists of an ensemble of regression trees that use the nonparametric Nelson-Aalen estimator for estimating the hazard function. For each tree, a subsample consisting of two-thirds of the training set was drawn (with replacement). In addition, for

each tree, a subset of the variables was randomly selected to be in the set of candidate variables to build the tree. To find the optimal hyperparameters, we performed a 5-fold cross-validation grid search using the training set only, which maximizes the C-index.

We used the functions GridSearchCV (scikit-learn, version 1.0.2) and concordance_index_censored from scikit-survival, version 0.15.1. We searched the hyperparameter space, which comprised number of trees (start = 30, end = 300), max_features (start = 10, end = 20), max_depth (start = 2, end = 4), min_samples_split (start = 2, end = 150), and min_samples_leaf (start = 10, end = 110). The final optimum values defined by the cross-validation grid search were the following: number of trees = 40, max_depth = 3, min_samples_split = 10, min_samples_leaf = 90, and max_feature = 15.

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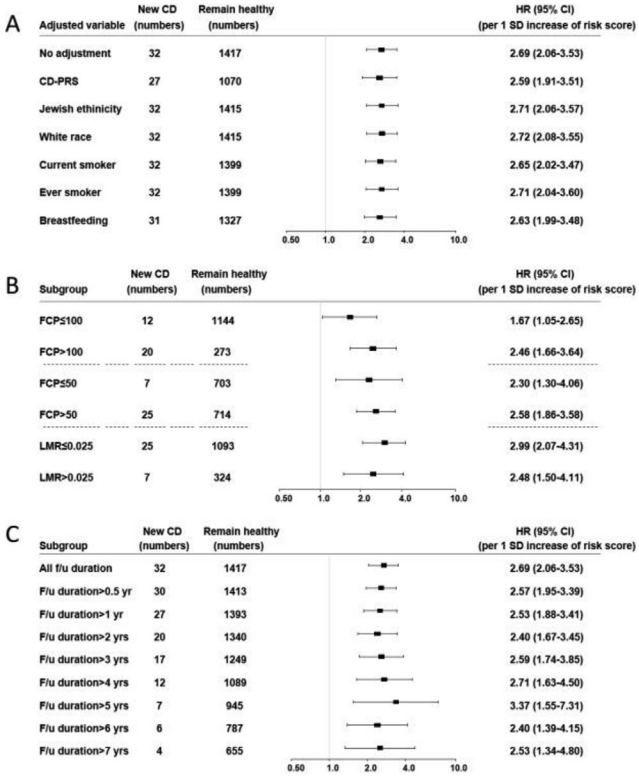
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Individual Variables	C-index (standard error)
GEM-IRS	0.789 (se 0.039)
LMR	0.519 (se 0.055)
FCP	0.759 (se 0.050)
Superpathway.of.fucose.and.rhamnose.degradation	0.615 (se 0.047)
NAD.biosynthesis.I..from.aspartate.	0.545 (se 0.059)
GDP.D.glycero..alpha..D.manno.heptose.biosynthesis	0.523 (se 0.055)
Fucose.degradation	0.631 (se 0.045)
F__Erysipelotrichaceae.g__Holdemania	0.579 (se 0.055)
Pantothenate.and.coenzyme.A.biosynthesis.I	0.571 (se 0.049)
F__Peptostreptococcaceae.g__Romboutsia	0.569 (se 0.056)
F__Lachnospiraceae.g__Lachnospira	0.643 (se 0.050)
F__Gemellaceae.g__Gemella	0.557 (se 0.028)
F__Lachnospiraceae.g__GCA.900066575	0.594 (se 0.044)
F__Ruminococcaceae.g__Angelakisella	0.579 (se 0.044)
Queuosine.biosynthesis	0.588 (se 0.056)
ADP.L.glycero..beta..D.manno.heptose.biosynthesis	0.496 (se 0.053)
Fatty.acid.salvage	0.459 (se 0.040)
F__Lachnospiraceae.g__Lachnospiraceae_ND3007_group	0.573 (se 0.058)
F__Clostridia_vadinBB60_group.g__Clostridia_vadinBB60_group	0.583 (se 0.042)
Superpathway.of.tetrahydrofolate.biosynthesis	0.554 (se 0.051)
F__Ruminococcaceae.g__.Eubacterium._siraeum_group	0.516 (se 0.050)
F__Lactobacillaceae.g__Lactobacillus	0.538 (se 0.034)
X6.hydroxymethyl.dihydropterin.diphosphate.biosynthesis.III..Chlamydia.	0.566 (se 0.050)
F__Lachnospiraceae.g__.Ruminococcus._torques_group	0.560 (se 0.068)
Superpathway.of.L.phenylalanine.biosynthesis	0.477 (se 0.051)
L.1.2.propanediol.degradation	0.613 (se 0.052)
Age	0.515 (se 0.050)
Sex	0.563 (se 0.043)
Multiplex	0.558 (se 0.036)
Current smoker	0.520 (se 0.003)
Ever smoker	0.512 (se 0.031)
CD-polygenic risk score	0.509 (se 0.053)
Breastfeeding	0.513 (se 0.045)
White race	0.531 (se 0.004)
Jewish	0.519 (se 0.033)

Supplementary Figure 1. Robustness of GEM-IRS hazard ratio (HR) after adjusting for confounders and in subgroup analyses in the pooled testing cohort. (*A*) Shown is the HR of the GEM-IRS on time to CD diagnosis after adjustment of a list of potential confounding effects individually. (*B*) Cox proportional hazards regression was used to identify the HR per 1 SD of GEM-IRS and 95% CI. Subgroup analysis based on FCP and LMR values is shown. (*C*) Subgroup analysis based on follow-up duration is shown. The *vertical line* indicates an HR of 1. (*D*) Predictive performance (C-index and SE) of the GEM-IRS along with that of the individual variables in the pooled testing cohort. f__, family level; F/u, follow up-time; g__, genus level. Crohn's disease polygenic risk score (CD-PRS) was a continuous variable; other variables were dichotomous variables (yes vs no). Ethnicity was asked by the question: "What is subject's cultural/ethnic background?" Ever smoking was asked by the question: "Have you ever smoked regularly, at least 1 cigarette per day, for 3 or more months?" Current smoking was asked by the question: "Are you currently smoking at least 1 cigarette per day?" Breastfeeding was asked by the question: "Were you breastfed as a baby exclusively or nearly exclusively for longer than 4 months?"

Supplementary Table 1. Baseline Characteristics of the Participants, According to Cohort[a]

Characteristic	North American training cohort (n = 1170)	Pooled testing cohorts		All cohorts combined (n = 2619)
		North American (n = 1141)	Israeli (n = 308)	
Age at recruitment, *y*, median (IQR)	18.0 (12.6 to 25.6)	18.0 (12.7 to 25.1)	16.2 (11.6 to 22.5)	17.7 (12.4 to 25.1)
Male sex, n (%)	552 (47.2)	532 (46.6)	150 (48.7)	1234 (47.1)
Relation to proband, n (%)				
Sibling	798 (68.2)	811 (71.1)	256 (83.1)	1865 (71.2)
Offspring	372 (31.8)	330 (28.9)	52 (16.9)	754 (28.8)
CD-multiplex family	86 (7.4)	93 (8.2)	24 (7.8)	203 (7.8)
Country of recruitment, n (%)				
Canada	992 (84.8)	941 (82.5)	—	1933 (73.8)
United States	178 (15.2)	200 (17.5)	—	378 (14.4)
Israel	—	—	308 (100)	308 (11.8)
Follow-up time, *y*, median (IQR)	7.3 (4.4 to 9.9)	7.0 (4.1 to 10.0)	5.3 (4.1 to 7.2)	6.8 (4.2 to 9.6)
FCP, *μg/g*, median (IQR)	53.8 (33.1 to 88.5)	53.2 (32.4 to 89.1)	43.4 (30.0 to 76.6)	52.4 (31.7 to 87.6)
LMR, median (IQR)	0.0179 (0.0134 to 0.0248)	0.0175 (0.0133 to 0.0236)	0.0197 (0.0145 to 0.0263)	0.018 (0.0135 to 0.0244)
CD-PRS, median (IQR)	–53.7 (–54.4 to –53.0)	–53.6 (–54.2 to –53.0)	–53.5 (–54.2 to –52.8)	–53.6 (–54.3 to –53.0)
New CD diagnosis,[b] n (%)				
Remained healthy	1141 (97.5)	1114 (97.6)	303 (98.4)	2558 (97.7)
Developed CD	29 (2.5)	27 (2.4)	5 (1.6)	61 (2.3)

NOTE. Continuous values are expressed as median (IQR) and categorical values are expressed as n (%). Percentages are proportion calculated within a given cohort.
CD-PRS, Crohn's disease polygenic risk score; IQR, interquartile range.
[a]No significant difference between randomly split North America training and testing cohorts was observed.
[b]Proportion of FDRs remaining healthy and of FDRs who developed CD as of February 28, 2021 at end of study.